

Geospatial Foundation Models Benchmarking

Datasets and Evaluation Challenges

Stefano Infusini

MSc Computer Science – Artificial Intelligence

University of Genoa – DIBRIS

September 2026

1 Current State of GFM Benchmarking Literature

Corley et al. [1] audited 152 papers related to Geospatial Foundation Models (GFMs) and argue that the current literature is substantially less comparable and reproducible than individual papers may suggest. Their study provides a systematic snapshot of the current GFM evaluation landscape.

The corpus also highlights some ambiguity in the use of the term “foundation model”. Only 46% of the 152 papers explicitly describe their proposed model as a foundation model in the title, abstract, or contributions. The remaining papers are earlier self-supervised remote-sensing models that previous GFM surveys already consider part of the same research lineage. The authors therefore treat the 46% figure as descriptive rather than as a criterion for inclusion.

The corpus was seeded from previous GFM surveys and expanded through two additional searches: an OpenAlex and Semantic Scholar citation-graph expansion, aimed mainly at recovering 2024–2025 papers not covered by the surveys, and a keyword search over remote-sensing self-supervised learning papers published between 2019 and 2023, before “foundation model” became common terminology in the field.

The resulting corpus contains 152 papers published between 2019 and 2025. Papers from 2026 were excluded because the year was incomplete at the time of submission. Metadata extraction was LLM-assisted and subsequently manually validated by the authors [1].

1.1 Model Weights Are Often Unavailable

Across the corpus, 39% of papers release no model weights [1]. A further 19% provide a public code repository but no released model artifact, meaning that reuse would require attempting to reproduce the original pretraining procedure.

This represents a basic obstacle to reproducible benchmarking. Before a model can be evaluated on a common downstream benchmark or used as a rerun baseline, the corresponding pretrained checkpoint must first be available.

1.2 The Field Does Not Converge on a Shared Core of Benchmarks

The 152 papers report evaluations on 401 distinct benchmarks, corresponding to 1,046 evaluation experiments in total [1].

The three most frequently used benchmarks—EuroSAT [2], NWPU-RESISC45 [3], and AID [4]—account for only 10.6% of all evaluations. The remaining 89.4% is distributed across 398 benchmarks, most of which appear in only one or two papers.

The problem is not simply that some benchmarks are more popular than others. Rather, individual papers frequently evaluate on almost disjoint benchmark sets.

For each paper, Corley et al. [1] compute the fraction of its downstream benchmarks that overlap with the ten most commonly used benchmarks in the corpus. The mean overlap is only 0.27, and 50 out of 143 papers, approximately 35%, have no overlap at all with the top ten.

At the corpus level, benchmark usage has a Gini coefficient of approximately 0.51. Moreover, this coefficient remains approximately stable after 2023, while the number of benchmarks appearing in only one paper increased from 13 in 2022 to 98 in 2025. The evidence therefore does not suggest convergence towards a shared evaluation core.

This creates a GFM-specific *benchmark lottery*: because models are rarely evaluated on sufficiently overlapping benchmark sets under common protocols, the published literature does not provide enough evidence to construct a reliable field-wide ranking of GFMs [1].

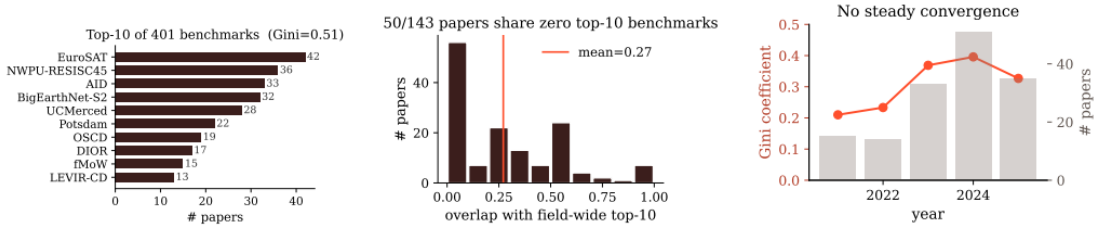


Figure 1: Benchmark usage across the audited GFM literature. The figure reports the most frequently used downstream benchmarks, per-paper overlap with the field’s top-10 benchmarks, and the evolution of benchmark concentration over time. Adapted from Corley et al. [1].

1.3 Reported Metrics Can Diverge Substantially Under the Same Nominal Protocol

The authors extracted every (model, benchmark, metric, evaluation protocol, training regime) tuple from the corpus, obtaining 10,817 results after benchmark and metric normalisation [1].

Evaluation protocols were grouped into categories such as fine-tuning, linear probing, kNN probing, zero-shot, and few-shot evaluation.

After filtering, 301 model–benchmark–protocol combinations appeared in at least two different papers. For each combination, the authors compute

$$\text{spread} = \max(\text{reported score}) - \min(\text{reported score}).$$

Of these 301 combinations, 76 have a spread of at least 5 percentage points, 46 at least 10 points, and 20 at least 20 points [1].

The largest discrepancy concerns Scale-MAE [5] evaluated on NWPU-RESISC45 using a linear probe. One paper reports an accuracy of 33.0, whereas the original Scale-MAE paper reports 89.6, corresponding to a difference of 56.6 percentage points, despite the use of the same released ViT-L checkpoint and the same nominal evaluation protocol [1].

This example illustrates that the label “linear probe” does not uniquely specify an experiment. Hidden choices may include the optimiser, learning rate, number of epochs, image resizing or cropping, normalisation, data augmentation, representation layer, train/validation/test split, class subset, sensor channels, and other adaptation details.

A second example reported by the authors concerns GPT-4o on UCMerced under zero-shot evaluation, where reported accuracy ranges from 43.5 to 88.8 percentage points. The corresponding papers do not describe the prompting configuration in sufficient detail to explain the discrepancy [1].

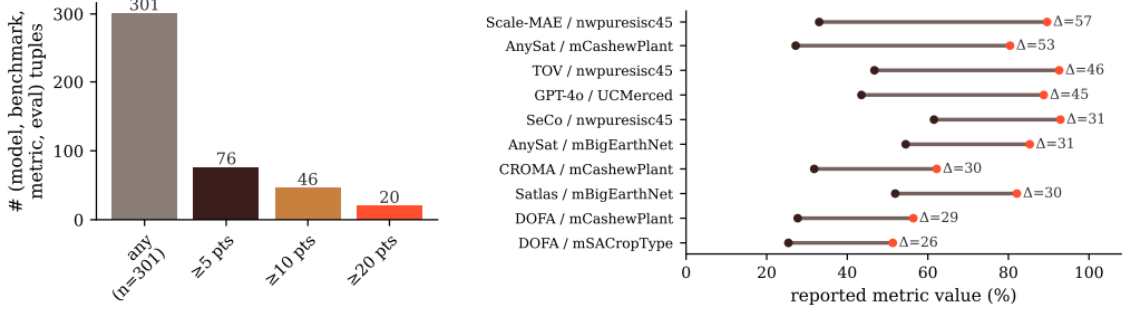


Figure 2: Largest cross-paper disagreements for repeated model–benchmark–protocol combinations. Even when the model, benchmark, and nominal evaluation protocol are the same, reported results can differ substantially. Adapted from Corley et al. [1].

Could This Be Explained by Random-Seed Variation?

The authors argue that random-seed variation alone is unlikely to explain the largest discrepancies.

Most repeated experiments agree closely, with a median spread close to zero, but the 90th-percentile spread is 12.7 percentage points [1].

Such a large tail is substantially greater than the run-to-run variation normally expected from simple classification heads under otherwise fixed conditions. The authors therefore interpret the largest disagreements primarily as evidence of hidden protocol differences.

1.4 Lack of a Shared Evaluation Harness

The geospatial community already provides benchmark collections and software infrastructure such as GEO-Bench [6], PANGAEA [7], and TorchGeo [8].

However, Corley et al. [1] argue that these resources do not yet constitute a community-wide evaluation harness comparable to those used in more mature evaluation ecosystems.

A benchmark collection may provide datasets, predefined splits, dataloaders, and example training scripts while still leaving substantial choices to the researcher. Consequently, two papers may both state that they evaluated on the same benchmark while actually executing different downstream experiments.

A stricter evaluation harness would additionally specify versioned task definitions, reference evaluation protocols, metrics, and automated checks, allowing a result to be reproduced from a model identifier and a task configuration.

Conceptually,

model → shared evaluation harness → standardised result.

Such a harness can reduce protocol-related degrees of freedom, but it cannot guarantee that the benchmark itself measures the intended deployment capability.

In particular, spatial dependence between training and test samples can inflate remote-sensing accuracy estimates [9, 10]. A perfectly reproducible evaluation may therefore still provide a misleading estimate of geographic generalisation.

Standardisation should also not necessarily imply forcing numerically identical hyperparameters on every architecture. Different models may require different optimisation settings. A fair comparison may instead require a common validation procedure and a comparable hyperparameter-search or compute budget, while allowing the final selected hyperparameters to differ between models.

1.5 Architecture and Pretraining Data Are Confounded

Another source of ambiguity arises when a new model simultaneously changes its architecture, pretraining data, and learning algorithm.

If several factors change at once, the observed downstream improvement cannot be attributed to one of them in isolation [1].

This is not an argument for requiring all GFMs to use identical pretraining data. Different, larger, private, or more diverse datasets may themselves be important contributions.

The problem arises when a paper attributes an improvement to an architectural or algorithmic modification without providing a controlled experiment in which the pretraining data are held fixed.

Corley et al. [11] show that apparent advantages of pretrained remote-sensing models over supervised ImageNet baselines can shrink substantially once preprocessing and distributional factors are controlled. The *No One Knows* study uses this as evidence that architecture and pretraining-data effects can be difficult to disentangle [1].

Similarly, Kaur et al. [12] study the effect of pretraining-data composition and show that changing the pretraining dataset alone can shift downstream performance by margins comparable to those often attributed to architectural novelty.

The corpus-level statistics make this comparability problem particularly clear.

Among the 126 papers for which a specific pretraining dataset could be extracted, the authors identify 87 distinct named primary pretraining datasets. Only 32 of the 126 papers, approximately 25%, share their complete pretraining configuration with at least one other paper. The remaining 94 use a configuration not shared by any other work in the corpus [1].

Consequently, comparisons in the GFM literature often involve systems that differ simultaneously in architecture, pretraining data, training objective, and downstream evaluation protocol.

Such comparisons remain useful at the *system level*, but they cannot by themselves establish which individual design choice caused a performance improvement.

1.6 Recommendations

Corley et al. [1] propose six recommendations aimed at improving comparability across the field:

1. **Release reusable model weights**, or explicitly state why they cannot be released.
2. **Evaluate on a minimum shared core of benchmark tasks**, while adding task-specific evaluations when needed.
3. **Distinguish copied and independently rerun baseline results**, including the corresponding source or configuration.
4. **Report uncertainty or repeated-run variability** when computationally feasible.
5. **Develop a shared third-party evaluation harness** with versioned protocols, common task definitions, and automated checks.
6. **Disentangle pretraining data, architecture, and learning algorithm** through controlled comparisons when causal claims are made.

2 Benchmark Datasets

2.1 Dataset Availability

Several efforts have already been made to catalogue, organise, and provide access to Earth Observation datasets for machine learning applications.

Schmitt et al. [13] provide a broad review of datasets developed for deep learning in Earth Observation, discussing their evolution across different sensors, tasks, spatial resolutions, and application domains. The same work introduces the IEEE GRSS Earth Observation Database (EOD), a community-driven catalogue intended to centralise metadata about publicly available remote-sensing datasets.

The EOD [14] provides an interactive search interface through which datasets can be filtered according to properties such as sensor modality, task, geographic location, dataset name, and keywords. This makes it a useful starting point for identifying candidate datasets satisfying specific experimental requirements.

A complementary effort is EarthNets [15], which reviews more than 500 publicly available remote-sensing datasets and organises them according to attributes including research domain, resolution, size, and relationships between datasets. Based on these attributes, the authors additionally select five large-scale datasets to form a benchmark and release the EarthNets platform for consistent model evaluation.

TorchGeo [8] provides a different but particularly useful resource from an experimental perspective. Rather than acting mainly as a dataset catalogue, TorchGeo implements a large collection of geospatial datasets through a common PyTorch interface, including benchmark datasets for classification, regression, semantic segmentation, object detection, instance segmentation, time-series analysis, and change detection. Many datasets can therefore be downloaded, loaded, and integrated into training pipelines with substantially less dataset-specific preprocessing code. This makes TorchGeo particularly relevant when selecting datasets that must later be evaluated across multiple models under a common experimental pipeline.

Additional community-maintained suites include `awesome-satellite-imagery-datasets` [16] and the `satellite-image-deep-learning/datasets` repository [17]. These resources are primarily useful for dataset discovery rather than as standardised benchmarking frameworks.

2.2 Consolidated Dataset Inventory

Index	Dataset	Year	Task	# Cls.	Sensor / modality	# Samples	Size (px)	Resolution (m)	GB
1	Indian Pines	2000	C	16	HS (Aerial)	1	145	20m	0.006
2	Salinas	2000	C	16	HS (Aerial)	1	365	3.7m	0.027
3	Botswana	2005	C	14	HS (Aerial)	1	875	30m	0.079
4	Kennedy Space Center	2005	C	13	HS (Aerial)	1	550	0.18m	0.057
5	Pavia Center	2009	C	9	HS (Aerial)	1	1096	–	0.124
6	Pavia University	2009	C	9	HS (Aerial)	1	610	–	0.033
7	UC Merced	2010	C	21	RGB	2.1K	256	0.3m	0.317
8	WHU-RS19	2012	C	19	RGB	1K	600	0.5m	0.1
9	Brazilian Coffee Scenes (BCS)	2015	C	2	RGB	2.9K	64	20m	0.004
10	RSSCN7	2015	C	7	RGB	2.8K	400	–	0.086
11	SAT-4	2015	C	4	RGB-NIR	500K	28	1~6m	1.15
12	SAT-6	2015	C	6	RGB-NIR	405K	28	1m	1.15
13	VEDAI	2015	C	9	IRGB	1.2K	1024	0.125m	3.99
14	Brazilian Cerrado-Savanna Scenes (BCSS)	2016	C	4	MS	1.3K	64	5m	0.011
15	NWPU-RESISC45	2016	C	45	RGB	31.5K	256	0.2~30m	0.405
16	RSC11	2016	C	11	RGB	1.2K	512	–	–
17	SIRI-WHU	2016	C	12	RGB	2.4K	200	2m	0.7
18	AID	2017	C	30	RGB	10K	600	3m	2.44
19	Austin Zoning	2017	C	5	RGB	3.7K	773x961	–	0.596
20	Kag. Planet Forest	2017	C	12	OPT (Satellite)	150K	256	–	33
21	MUSIC4GC (Golf Course)	2017	C	2	MS+L	83.4K	16	30m	0.37
22	RSD46-WHU	2017	C	46	RGB	117K	256	0.5~2m	–
23	RSI-CB128	2017	C	45	RGB	36K	128	0.3~3m	0.879
24	RSI-CB256	2017	C	35	RGB	24K	256	0.3~3m	2.24
25	Ships in Satellite Imagery	2017	C	2	RGB	4K	80	3m	0.343
26	Solar Power Plants MUSIC4P3	2017	C	2	MS+L	1.3M	16	30m	4.6
27	TiSeLaC	2017	C	9	RGB-NIR+MT	23	2866x2633	30m	–
28	Cactus Aerial Photos	2018	C	2	RGB	24K	32	–	0.053
29	EuroSAT	2018	C	10	S2	27Kx13	64	10m	1.92
30	FMoW	2018	C	63	RGB+MS	523.8K	–	0.3m	3.5K
31	Gaofen Image (GID)	2018	C	15	RGB-NIR	150	7200	4m	47
32	MASATI-v2	2018	C	7	RGB	7.4K	512	0.08~2m	2.3
33	MSLCC	2018	C	4	SAR+MS	2	31~957	10m	0.5
34	PatternNet	2018	C	38	RGB	30.4K	256	0.062~4.693m	1.3
35	SanFranBay Ships Planet	2018	C	1	OPT (Satellite)	4K	80	–	0.194
36	xView	2018	C	60	RGB+MS	1.4K	3000	0.3m	20
37	AIDER	2019	C	4	OPT (Drone)	2.6K	240	–	0.275
38	BigEarthNet-MM	2019	C	19	S1/S2	590.3K	20~120	1~6	121
39	DSCR	2019	C	7	RGB	20.7K	150~800	–	–
40	HistAerial	2019	C	7	Grayscale	42K	25~100	–	7.6
41	ICONES-HSI	2019	C	9	HS (Aerial)	468x224	300	–	11.5
42	Kag. Find a Car Park	2019	C	2	RGB	3.3K	1296	–	2.75
43	Kaggle Cloud Detection	2019	C	4	RGB	9.2K	1750	–	6
44	Kag. Satellite Hurricane Damage	2019	C	–	RGB	16K	128	1m	0.064
45	OPTIMAL-31	2019	C	31	RGB	1.9K	256	–	0.024
46	Ship Classification	2019	C	7	OPT (Satellite)	2K	–	–	–
47	SloveniaLandCover	2019	C	10	MS	940	–	–	–
48	So2Sat LCZ42	2019	C	–	S1+S2	400.7K	32	10m	51.8
49	USTC SmokeRS	2019	C	6	RGB	6.2K	256	1000m	0.795
50	WiDS Datathon 2019	2019	C	2	RGB	11K	256	3m	0.46
51	Airbus Wind Turbines Patches	2020	C	1	RGB+MS	358K	128	1.5m	1
52	AiRound	2020	C	11	Multi	11.8K	300	–	33
53	Bijie Landslide	2020	C	1	RGB	2.8K	200	0.68m	0.513
54	BrazilDAM	2020	C	2	RGB	769	384	10~60m	57
55	BreizhCrops	2020	C	9	S2+MT	610K	–	10~60m	–
56	CaneSat	2020	C	2	S2+MT	1.6K	10	10m	0.006
57	CLRS	2020	C	25	RGB	15K	256	0.26~8.85m	1.74
58	CV4A Kenya	2020	C	7	S2	8.8K	2500	10m	3.5
59	CV-BrCT	2020	C	9	RGB	24K	500	–	9.2
60	DND-SB	2020	C	7	OPT (Drone)	5.6K	7296	–	49
61	LandCoverNet Af	2020	C	7	Multiple (Satellite)	2K	256	–	632
62	LandCoverNet As	2020	C	7	Multiple (Satellite)	2.8K	256	–	610
63	LandCoverNet Au	2020	C	7	Multiple (Satellite)	600	256	–	126
64	LandCoverNet Eu	2020	C	7	Multiple (Satellite)	840	256	–	249
65	LandCoverNet NA	2020	C	7	Multiple (Satellite)	1.6K	256	–	351
66	LandCoverNet SA	2020	C	7	Multiple (Satellite)	1.2K	256	–	215
67	MAFAT Challenge	2020	C	–	SAR	–	–	–	–
68	MLRSNet	2020	C	46	RGB	109.2K	256	0.1~10m	1.25
69	OVERHEAD MNIST	2020	C	9	Grayscale	1K	28	–	0.018
70	Rwanda Crop Type	2020	C	6	OPT (Drone)	6.5K	200	–	56
71	SAR-based Sea Ice Dataset	2020	C	6	SAR (Satellite)	98.8K	28	–	0.14
72	Sen1Floods11	2020	C	1	S1	4.8K	512	10m	14.3
73	SenseEarth classify	2020	C	51	RGB	70K	100~12655	0.2~153m	10.8
74	Urban Tree Detection Data	2020	C	2	RGB-NIR	60	256	0.6m	0.114
75	2019 Mali CropType	2021	C	4	MS (Satellite)	33*13*41	256	–	0.456
76	CropHarvest	2021	C	–	Multi	95.2K	–	10~60m	22.7
77	CropRotation	2021	C	–	MS (Satellite)	–	–	–	53.4
78	DENETHOR	2021	C	9	S1+S2	2*(365 + 60*14)	8000	3m	254
79	FGSCR-42	2021	C	42	RGB	9.3K	500	–	–
80	MAFAT-Fine-Grained	2021	C	37	RGB	4.2K	–	0.05m-0.15m	–
81	Million AID	2021	C	28	RGB	1M	150~550	0.5~153m	134
82	MultiScene	2021	C	36	RGB	100K	512	0.3~0.6m	6
83	NaSC-TG2	2021	C	10	RGB-NIR	20K	256	100m	–
84	Satellite Image Classification	2021	C	4	RGB	5.6K	256	–	0.023
85	SeaIceClassification	2021	C	8	SAR (Satellite)	6.4K	256	–	2.9
86	South Africa Crop Type	2021	C	9	S1+S2	122.7K	–	–	82.8
87	TG1HRSSC	2021	C	9	HS	204	512	5m-20m	0.277
88	Urban Village S2UC	2021	C	2	RGB	1.7K	224	2m	1.8
89	ZueriCrop	2021	C	48	S2	28Kx71	24	10m	38.5
90	ADVANCE	2022	C	13	OPT (Satellite)	5.1K	256	–	4.4
91	AgriFieldNet	2022	C	13	MS (Satellite)	1.2Kx12	256	–	0.32

Continued on next page

Table 1 continued

Index	Dataset	Year	Task	# Cls.	Sensor / modality	# Samples	Size (px)	Resolution (m)	GB
92	Aircraft SAR-ACD	2022	C	20	SAR	4.3K	64	–	–
93	AnthroProtect	2022	C	2	OPT (Satellite)	23.9K	256	–	18.1
94	ARGO	2022	C	2	MS (Satellite)	1.8K	1400	–	14.3
95	Canadian Cropland	2022	C	10	S2	78.5K	64	10m	26
96	MUSIC4HA (Hot Area)	2022	C	6	S2	2.5K	16	10m	0.01
97	OMBRIA	2022	C	2	S1+S2	3.4K	256	10m-20m	0.19
98	S2-Agri	2022	C	–	MS (Satellite)	–	32	–	10.8
99	Space2Ground	2022	C	2	Multi	10.1K	260	10~60m	0.501
100	SynthWakeSAR	2022	C	10	SAR	46.1K	96000	3.3m	4.3
101	The Auto Arborist Dataset	2022	C	344	RGB+MS	2.6M	1~24	–	24
102	TreeSatAI	2022	C	47	Multi	50.4K	304	0~1	16.3
103	UAE-RS	2022	C	–	OPT	6.1K	–	–	–
104	XAI4SAR	2022	C	–	SAR (Satellite)	5.9K	256	–	0.691
105	Digital Typhoon	2023	C,R	8	IR	189.4K	512	5000m	–
106	FireRisk	2023	C	7	OPT (Satellite)	91.9K	–	–	14
107	MAR20	2023	C	20	RGB	3.8K	800×800	1m	1.1
108	m-EuroSAT (GEO-Bench)	2023	C	10	S2 MS	27K	64	10~60m	1.2
109	EuroCropsML	2024	C	176	S2 MS	706.7K	–	10m	–
110	QuakeSet	2024	C,R	2	S1 SAR	3.3K	512	10m	11
111	TalloS	2024	C	1160	Multi	335K	–	10m	–
112	TreeSatAI-TS	2024	C	20	Multi	–	–	0.2m-10m	30
113	BigEarthNetV2-S2 (Copernicus-Bench)	2025	C	19	S2 MS	24K	120	10m	–
114	GEO-Bench-2	2025	C,S,R,OD,I	–	Multi	19 datasets	–	–	–
115	Landsat e-Reporting	2025	C	–	Landsat MS	3 datasets	–	30m	–
116	MONITRS	2025	C,IC,C,S,	–	S2 RGB+Text	10K	512	10m	6.39
117	REOBench	2025	OD,IC	–	OPT+Text	6 subsets	–	–	–
118	CLearRS	2026	C,OD	–	Multi	7 datasets	–	–	–
119	SwissCrop25	2026	C,T	73	S2 MS	7 seasons	–	10m	–
120	Earth Surface Temperature	2017	R	–	Climate variables	16	–	–	0.09
121	Greenhouse Gas	2017	R	–	Gas emission	–	–	–	0
122	Forest Canopy Height	2018	R	–	LiDAR	1.1K	–	1m	62.9
123	TenGeoP-SARwv	2019	R	10	SAR	37K	–	5m	31.7
124	Poverty in Africa	2020	R	–	RGB	32.8K	256	–	5.58
125	Tropical Cyclone	2020	R	–	GOES 8–16 MSI	108.1K	256×256	4K–8K	–
126	Air Quality e-Reporting	2021	R	–	–	–	–	–	–
127	Forel-Ule Index global inland waters	2021	R	–	MS	–	–	500m	0.004
128	Hurricane Wind Speed	2021	R	–	MS	114.6K	–	–	–
129	SustainBench Crop Yield	2021	R	–	MODIS MSI	11K	32	–	–
130	USAVars	2021	R	–	NAIP RGB+NIR	100K	–	4m	–
131	Hyperview Challenge	2022	R	–	HS	2.9K	11~250	2m	1.85
132	So2Sat Population	2022	R	–	DEM+S2	–	–	10m	–
133	BioMassters	2023	R	–	S1/S2	13K	256	10m	201
134	SKIPPD	2023	R	–	RGB	363.4K	64	–	–
135	PhilEO Bench	2024	R,S,C	–	S2 MS	–	–	10m	400
136	SolarCube	2024	R,T	–	Multi	–	–	–	–
137	Western USA Live Fuel Moisture	2024	R	–	L8+S1	2.6K	–	–	–
138	CropClimateX	2026	R,C,T	–	Multi	15.5K cubes	–	–	6.3K
139	YieldSAT	2026	R,T	–	Multi	12.2M	–	10m	–
140	Air Quality	–	R	–	Multi-sensor	9.4K	–	–	–
141	DFC07	2007	S	–	Multiple (Satellite)	1	787	–	0.019
142	DFC08	2008	S	5	HS (Aerial)	–	–	–	0.194
143	ISPRS 2D - Potsdam	2011	S	6	RGB+nDSM	38	6000	0.05m	16
144	ISPRS 2D - Vaihingen	2011	S	6	RGB+nDSM	33	2200	0.09m	17
145	Aerial Image Segmentation	2013	S	–	RGB	80	512	0.3~1m	0.007
146	DFC13	2013	S	16	Multiple (Aerial)	145	–	–	1
147	Massachusetts Roads	2013	S	–	RGB	1.2K	1500	1m	7.55
148	DFC14	2014	S	7	Multiple (Aerial)	–	–	–	–
149	DFC15	2015	S	–	Multiple (Aerial)	7	10000	–	–
150	DFC2015 Zeebrugues	2015	S	8	Multi	7	0~100	0.05m	0.002
151	ERM PAIW	2015	S	2	RGB	41	4000	0.3m	0.635
152	Zurich Summer	2015	S	–	RGB-NIR	20	1000	0.61m	0.132
153	Chikusei	2016	S	19	HS (Aerial)	1×128	2517	–	1.4
154	DFC16	2016	S	–	OPT	–	–	–	–
155	HD-Maps	2016	S	5	RGB	20	4000	0.3m	0.146
156	L8 Biome	2016	S	5	L8 MSI	96	8.9K	15~30m	96
157	SpaceNet-1	2016	S	1	RGB+MS	9.7K	650	0.5~1m	31
158	AIS From Online Maps	2017	S	2	RGB	1.7K	3000	0.5m	23.8
159	CITY-OSM	2017	S	–	OPT (Aerial)	21	2500	–	23.8
160	DFC17	2017	S	17	Multiple (Multiple)	–	–	–	–
161	EvLab-SS	2017	S	11	RGB	60	4500	0~0.2	–
162	GTA_BEV_SVA	2017	S	–	OPT	1M	–	–	–
163	INRIA	2017	S	1	RGB	360	1500	0.3m	19.5
164	Inria Aerial Image Labeling	2017	S	2	RGB	360	5000	0.3m	–
165	SpaceNet-2	2017	S	1	RGB+MS	24.6K	650	0.3m	182
166	SpaceNet-3	2017	S	1	Multi	3.7K	1300	0.3~1.24m	182
167	Urban 3D Challenge	2017	S	2	RGB	157K	–	0.5m	–
168	38-Cloud	2018	S	1	RGB	17.6K	384	30m	16
169	AerialLanes18	2018	S	–	RGB	20	5616	0.125m	–
170	Aeroscapes	2018	S	11	RGB	3.3K	1280	–	0.752
171	Agricultural Crop Cover	2018	S	2	MS	40	–	30m	4.4
172	DeepGlobe-Bldng	2018	S,C	1	OPT (Satellite)	24.6K	650	–	–
173	DeepGlobe-LandCover	2018	S	7	RGB	1.1K	2448	0.5m	2.75
174	DeepGlobe-Road	2018	S	1	OPT (Satellite)	8.6K	1024	–	3.84
175	DFC18	2018	S	20	Multi	1	–	0.05~1m	19.8
176	DLRSD	2018	S	21	RGB	2.1K	256	0.3m	0.004
177	HyRANK	2018	S	14	HS	5	0~1	30m	0.4
178	RIT-18	2018	S	–	MS	1	9300	0.047m	1.5
179	RoadNet	2018	S	2	RGB	20	–	0.21m	0.905
180	SEN1-2	2018	S	–	Multiple (Satellite)	282.4K	256	–	43.7

Continued on next page

Table 1 continued

Index	Dataset	Year	Task	# Cls.	Sensor / modality	# Samples	Size (px)	Resolution (m)	GB
181	SynthAer	2018	S	8	RGB	765	1280	–	1
182	Urban Drone Dataset (UDD)	2018	S	6	RGB	301	4~96	–	1.1
183	WHDLD	2018	S	6	RGB	4.9K	256	2m	0.102
184	AeroRIT	2019	S	–	RGB+HS	1	3900	0.4m	1.8
185	AIRS	2019	S	1	RGB	1K	10000	0.075m	17.6
186	Chesapeake Land Cover	2019	S	16	RGB+MS	250K	240	1m	140
187	DLR-SkyScapes	2019	S	31	RGB	16	4680	0.13m	–
188	DroneDeploy	2019	S	7	RGB	55	0~6	0.1m	–
189	GF2 Dataset for 3DFGC	2019	S	5	RGB-NIR	11	2~652	4m	0.056
190	HRC_WHU	2019	S	2	RGB	150	1280×720	0.5~15m	0.17
191	ManipalUAVid	2019	S	4	OPT (Drone)	711	1280	–	1.05
192	Open AI Challenge: Caribbean	2019	S	5	RGB	7	52~318	0.04m	–
193	PolSF	2019	S	6	SAR (Satellite)	5	–	–	–
194	Salient Objects ORSSD	2019	S	8	RGB	800	500	–	0.026
195	Semantic Drone	2019	S	20	RGB	400	5000	–	4
196	SEN12MS	2019	S,C	–	MS+SAR	180.7K	256	10m	510
197	SpaceNet-5	2019	S	1	Multi	2.4K	1300	0.3m	84.1
198	95-Cloud	2020	S	1	RGB	34.7K	384	30m	18
199	ActiveFireL8	2020	S	1	MS (Satellite)	2M×10	128	–	546
200	AIR-CD	2020	S	1	RGB-NIR	34	7300	4m	12
201	BDCI2020	2020	S	7	RGB	146K	256	2m	1.3
202	BHPools&WaterTanks	2020	S	1	OPT (Aerial)	350	3000	–	1.9
203	DFC20	2020	S	17	S1+S2	180.7K	–	10m	19.2
204	DubaiSemSeg	2020	S	6	OPT (Satellite)	72	800	–	0.032
205	EORSSD	2020	S	2	RGB	2K	500	–	0.06
206	FLAME	2020	S,C	2	MS (Drone)	48K	3840	–	43.7
207	FUSAR-Map	2020	S	4	SAR (Satellite)	610	1024	–	–
208	Ghana Crop Type	2020	S	5	Multi	2881*14*25	64	3~10m	312
209	GTA-V SID	2020	S	–	OPT (Satellite)	121	500	–	0.1
210	HKH Glacier Mapping	2020	S	2	MS (Satellite)	14.2K×15	512	–	29.4
211	Kaggle aerial segmentation	2020	S	6	RGB	72	800	–	0.033
212	Kaggle Buildings	2020	S	2	OPT (Satellite)	6K	256	–	0.878
213	Kaggle Water Bodies	2020	S	2	MS (Satellite)	2.8K	1000	–	0.274
214	Kag. Massachusetts Buildings	2020	S	1	RGB	151	1500	1m	3
215	Kenya Crop Type	2020	S	7	S2 MSI	4.7K	3035x2016	10m	–
216	LandCoverAI	2020	S	–	RGB	41	9000	0~0.2	1.4
217	Open Cities AI Challenge	2020	S	2	RGB	11K	1~24	0.03~0.2m	81.5
218	OpenSARUrban	2020	S	10	SAR (Satellite)	33.4K	100	–	2.2
219	Parking APKLOT	2020	S	2	RGB	501	–	–	3
220	Power TTPL	2020	S	3	RGB	1.1K	3840	–	4.2
221	Rwanda Built-Up	2020	S	1	OPT (Satellite)	787	256	–	0.07
222	S2 Cloud Mask Catalogue	2020	S,C	25	S2	513×13	1022	20m	15.4
223	SemCity Toulouse	2020	S	–	MS	16	3500	0.5~2m	5.2
224	SEN-12-FLOOD	2020	S	1	Multi	336	–	10m	12.2
225	South Sudan Crop Type	2020	S	5	Multiple (Satellite)	687*14*25	64	–	977
226	SpaceNet-6	2020	S	1	SAR+RGB	3.4K	900	0.5m	0.368
227	SpaceNet-7	2020	S	1	MS+MT	1.5K	1024	4m	20.6
228	Synthinel-1	2020	S	1	RGB	2.1K	572	0.3m	1
229	TimeSen2Crop	2020	S	16	S2	1M	10980	10m	1.1
230	TTPLA	2020	S	–	OPT (Aerial)	1.1K	3840	–	4.2
231	UAVid	2020	S	–	RGB	420	4000	–	5.88
232	WHU-Hi	2020	S	11	HS (Drone)	3	1217	0.109m	0.817
233	WHU-Hi-HongHu	2020	S	22	HS	1	940x475	0.043m	–
234	Arctic Sea Ice Masking	2021	S	8	RG-NIR	3.4K	357x306	10m	0.092
235	Benin Cashews	2021	S	6	S2 (labels corrected with Pléiades)	70	1186x1122	0~1	–
236	DFC21-DSE	2021	S	4	Multi	6K	16	10~750m	18
237	DoriaNet	2021	S	3	OPT (Drone)	2.4K	1280	–	0.067
238	ETCI2021	2021	S	2	SAR (Satellite)	66.8K	256	–	5.3
239	EuroCrops	2021	S	43	S2	805.4K	–	10m	8.6
240	FloatingObjects	2021	S	–	MS (Satellite)	–	–	–	15.4
241	FloodNet	2021	S	10	RGB	2.3K	4000	0.015m	12.1
242	LoveDA	2021	S	7	RGB	6K	1024	0.3m	9.6
243	MARIDA	2021	S	15	MS (Satellite)	1.4K×10	256	–	1.2
244	Mini Inria Aerial Image Labeling Dataset	2021	S	2	RGB	32.5K	512	0.3m	–
245	MUESLI	2021	S	41	Multiple (Aerial)	1×415	12800	–	37.7
246	MultimodalBen	2021	S	30	Multiple (Multiple)	568	680	–	0.489
247	PASTIS	2021	S	18	S2	2433*45*10	128	10m	29
248	PASTIS-R	2021	S	18	Multiple (Satellite)	2.4K	128	–	54
249	RescueNet	2021	S	12	OPT (Drone)	4.5K	3500	–	21.8
250	Rwanda Field Boundary	2021	S	2	RGB+NIR	70	256	4.7m	–
251	S2-SHIPS	2021	S	3	MS (Satellite)	16×16	1783	–	2.9
252	Sen4AgriNet	2021	S	–	S2	–	–	10~60m	262
253	SmallholderPlant	2021	S	6	MS (Satellite)	70×13	1180	–	1
254	The Azavea Cloud Dataset	2021	S	2	S2	32	–	10m-60m	–
255	UOPNOA	2021	S	–	Multiple (Multiple)	36K	–	–	9.7
256	AIR-PolSAR-Seg	2022	S	6	SAR	2K	512	8m	0.609
257	Burned Area Delineation	2022	S	–	Multiple (Satellite)	–	–	–	14
258	C2S-MS	2022	S	3	Multiple (Satellite)	1.8K	512	–	12
259	DFC2022	2022	S	15	RGB	4K	2000	0.5m	–
260	DFC22-SSL	2022	S	14	Multiple (Aerial)	5K	2000	–	42
261	FireCube	2022	S	1	Multiple (Multiple)	4.3K×58	700	–	23.5
262	Five-Billion-Pixels	2022	S	24	RGB+MS	150	7200×6800	4m	104
263	FLAIR #1	2022	S	13	OPT (Aerial)	77.4K	512	–	70
264	GeoNRW	2022	S	10	RGB+nDSM	7.8K×3	1000	1m	30
265	GrowliFlower	2022	S	4	MS (Drone)	8.4K	256	–	11.7
266	HOSD	2022	S	1	HS (Aerial)	18×224	1700	–	2.2
267	Landslide4Sense	2022	S	1	Multiple (Satellite)	4.8K×14	128	–	2.4
268	MapInWild	2022	S	11	Multiple (Satellite)	1K×8	1920	–	365
269	MDAS	2022	S	20	Multi	3	100~18K	0.3~30m	–
270	Microsoft RoadDetections	2022	S	1	RGB	20K	1088	1m	9.25

Continued on next page

Table 1 continued

Index	Dataset	Year	Task	# Cls.	Sensor / modality	# Samples	Size (px)	Resolution (m)	GB
271	MiniFrance	2022	S	12	RGB	2.1K	10000	0.5m	93.1
272	MMFlood	2022	S	1	SAR (Satellite)	1.7K×2	2000	–	11.3
273	MultiSenGE	2022	S	14	S1+S2	1.1M×12	256	10m	530
274	NASA-Harvest	2022	S	1	MS (Satellite)	70	256	–	0.3
275	NDWS	2022	S	1	Multiple (Multiple)	18.4K×12	64	–	2
276	OnCloudN	2022	S	1	OPT (Satellite)	22.7K	512	–	50
277	OpenEarthMap	2022	S	8	RGB	5K	1024	0.25m-0.5m	9.1
278	OpenSentinelMap	2022	S	15	RGB+S2	137K	192	10m-60m	445
279	Paddy Rice Labeling South Korea	2022	S	–	S2	–	–	–	0.002
280	Paddy Rice Maps South Korea	2022	S	–	S1	12.9K	256	10m	0.198
281	RDCME	2022	S	1	OPT (Satellite)	775	256	–	–
282	RID	2022	S	16	RGB	2K	–	0.1m	1.5
283	S12 ScotInd Cloud	2022	S	1	Multiple (Satellite)	–	–	–	0.96
284	Sat Burn Area	2022	S	1	MS (Satellite)	73×14	1500	–	14
285	SeasoNet	2022	S	33	S2	1.8M	120	10m	229
286	Sentinel-2 Cloud Cover	2022	S	2	MS	22.7K	–	10m-60m	51.2
287	SpaceNet-8	2022	S	2	PAN+RGB	1.2K×2	1200	0.3~0.8m	6.8
288	SyntCities	2022	S	13	OPT	16.2K	–	–	–
289	TimeSpec4LULC	2022	S	29	MS+MT	–	–	500m	60
290	UBC	2022	S	61	OPT (Satellite)	800	600	–	0.675
291	Urban Building Classification	2022	S	61	RGB	800	600	0.5~0.8m	0.675
292	WHU-OHS	2022	S	24	HS	7.8K	512	10m	94.9
293	WildfireDanger	2022	S	1	Multiple (Multiple)	–	–	–	17
294	A14SmallFarms	2023	S	2	S2 MS	62 tiles	–	10m	–
295	DFC23	2023	S,C	12	Multiple (Satellite)	–	–	–	–
296	FLAIR #2	2023	S	13	Multi	77.8K	512	0.2m-10m	–
297	HLS Burn Scars	2023	S	2	HLS	804	512	30m	–
298	I/O Bench	2023	S	–	Landsat MSI	1	8000	30m	–
299	Luojia-HSSR	2023	S	23	RGB	6.4K	256	0.75m	–
300	SSL4EO-L Benchmark	2023	S	17/134	Landsat MSI	25K	264	30m	–
301	Chesapeake Roads	2024	S	1	RGBN	–	–	1m	28.3
302	Fields Of The World	2024	S,I	2/3	S2 MSI	70.8K	256	10m	–
303	FUSU	2024	S,CD	17	S1/S2	>30B px	–	10m	–
304	GAMUS	2024	S	7	RGB+nDSM	–	–	1m	10
305	MADOS	2024	S	15	S2 MS	2.8K	40~240	10~60m	5.35
306	PASTIS-HD	2024	S,C,I	20	Multi	–	–	1m	–
307	PhilEO-downstream	2024	S,C,R	11	S2	–	256	10m	400
308	QuickQuakeBuildings	2024	S	2	RGB+SAR	–	–	0.3m	–
309	reBEN	2024	S,C	–	S1/S2	549.5K	20~120	10m-60m	–
310	SICKLE	2024	S,C	21	S1/S2+L8 TS	34.8K	–	3m-30m	–
311	Ticino	2024	S	10	Multi	–	–	1.86m-30m	–
312	UrbanSARFloods	2024	S	2	S1 SAR	8.9K	–	10m	–
313	WALD5Dplus	2024	S	–	RGB+SAR	–	–	10m	–
314	DL4GAM	2025	S	2	S2 MSI	2.3K/11.4K	256	10m	–
315	OpenEarthMap-SAR	2025	S	8	SAR+OPT	1.5M seg-ments	–	–	–
316	50 Cities	2026	S	–	S1/S2	50	–	10m	10.7
317	BELDE	2026	S	7	S2 MS	1.1M	–	10m	–
318	FLAIR-HUB	2026	S	19/23	Multi	241.1K	10~512	0.2~10m	–
319	CaFFe	–	S	2/4	SAR	19.1K	512	6~20m	–
320	EnviroAtlas	–	S	10	Multi	–	–	1m	–
321	L7 Irish	–	S	5	L7 MSI	206	7.5K~8.4K	15~30m	–
322	Solar Plants Brazil	–	S	2	RGB+NIR	272	256	10m	–
323	Substation	–	S	2	S2 MSI	27K	228	10m	–
324	CrowdAI Mapping	2018	I	1	RGB	401.8K	300	–	5.33
325	DeepGlobe (Building)	2018	I,S	2	Multi	24.6K	650	0.5m	–
326	iSAID	2020	I	15	RGB	2.8K	4000	–	6.54
327	GlobalMLBuildingFootprints	2024	I	2	Aerial RGB	–	–	0.5m	–
328	SEN12TP	2023	T	–	S1/S2	2.3K	–	–	–
329	DFC09	2009	CD	–	Multiple (Satellite)	2	–	–	0.098
330	SZTAKI AirChange	2012	CD	2	RGB	13	800	1.5m	0.042
331	Cross-sensor Bastrop	2015	CD	2	MS	4	0~153	3~12	–
332	AIST Building Change Detection	2017	CD	2	RGB	16.9K	160	0.4m	17.8
333	HSI CD	2017	CD	5	HS (Multiple)	3×224	800	–	1.7
334	AICD	2018	CD	–	OPT (Aerial)	1K	800	–	1.7
335	CDD (season-varying)	2018	CD	2	RGB	16K	256	0.03~0.1m	2.7
336	GETNET dataset	2018	CD	2	HS	1	463×241	30m	0.05
337	Hermiston City Oregon	2018	CD	5	HS	1	390×200	30m	–
338	Hyperspectral CD	2018	CD	5	HS	3	0~98	30m	1.7
339	OSCD	2018	CD	2	S2	24	600	10m	0.48
340	Synthetic & Real	2018	CD	–	OPT (Aerial)	16K	256	–	2.7
341	WHU Building change detection	2018	CD	2	RGB	2	15354×32507	0.075m	5
342	xView2	2018	CD	4	RGB	22.1K	1024	0.5m	51
343	Flood California flood dataset	2019	CD	2	RGB+MS	1	1534×808	5m,30m	0.33
344	HRSCD	2019	CD	7	RGB	582	10000	0.5m	5.06
345	MtS-WH	2019	CD	1	RGB-NIR	190	150	1m	0.459
346	DSIFN	2020	CD	1	RGB	394	512	10m	0.454
347	Google Data Set	2020	CD	2	RGB	19	1000~5000	0.55m	0.6
348	Hi-UCD	2020	CD	9	RGB	1.3K	1024	0.1m	–
349	LEVIR-CD	2020	CD	1	RGB	637	1024	0.5m	2.7
350	SECOND	2020	CD	30	RGB	4.7K	512	0.5~3m	2.2
351	Zhang et al. CD dataset	2020	CD	2	RGB+NIR	4	458~1.4K	2~5	0.1
352	DFC21-MSD	2021	CD	15	MS+MT	2.2K×9	4000	1~30m	325
353	Dynamic EarthNet Challenge	2021	CD	7	RGB	22.5K	1024	3m	–
354	HTCD	2021	CD	2	RGB	3.8K	256~2K	0.074m-0.597m	1.74
355	MultimodalSupCD	2021	CD	–	Multi	–	–	–	–
356	MUNO21	2021	CD	1	OPT (Aerial)	156	12000	–	14.3
357	RapidA14EO	2021	CD	44	MS+RGB	500K	–	3m,10m	–
358	Relative Radiometric Normalization	2021	CD	2	MS	7	300~5000	0~6	1
359	S2Looking	2021	CD	1	RGB	5K×2	1024	0.5~0.8m	10.2
360	S2MTCP	2021	CD	–	S2	1.5K	600	10m	10.6

Continued on next page

Table 1 continued

Index	Dataset	Year	Task	# Cls.	Sensor / modality	# Samples	Size (px)	Resolution (m)	GB
361	SYSU-CD	2021	CD	5	RGB	20K×2	256	0.5m	5.1
362	WH-MAVS	2021	CD	15	RGB	47.1K	200	1.2m	—
363	CDVQA	2022	CD	19	RGB	3K	512	0.5~3m	—
364	Cropland Change Detection (CLCD)	2022	CD	2	RGB	600	512	0.5m-2m	0.5
365	DynamicEarthNet	2022	CD,S	7	MS (Satellite)	54.8K	1024	—	524
366	Dynamic World	2022	CD	9	S2	—	—	10m	—
367	MSBC	2022	CD	2	Multi	3.8K	256	2m	3.9
368	MSOSCD	2022	CD	2	Multi	5.1K	256	10~60m	2.7
369	BANDON	2023	CD	2	RGB	2.3K	2048×2048	0.6m	3.4
370	BraDD-SITS	2023	CD,S	2	S1 TS	13.2K	—	10m	17.7
371	ChangeNet	2024	CD	6	Multi-temporal RGB	—	—	0.3m	—
372	BRIGHT	2025	CD	4	RGB+SAR	3.2K	1024	0.1~1m	—
373	FAIR-EO CD Benchmark	2026	CD	—	Multi	—	—	—	—
374	CaBuAr	—	CD	2	S2 MSI	424	512	20m	—
375	ChaBuD	—	CD	2	S2 MSI	356	512	10m	—
376	MSTAR	1996	OD	—	SAR (Satellite)	—	—	0.3m	—
377	SAR Vehicle	2005	OD	1	SAR (Aerial)	104	—	—	7.5
378	VIVID - PETS2005	2007	OD	1	OPT (Aerial)	9.3K	640	—	—
379	TaS	2008	OD	1	RGB	30	792	0.5m	0.97
380	OIRDS	2009	OD	5	RGB	900	256~640	0.15m	0.153
381	NWPU-VHR10	2014	OD	10	RGB+IRRG	800	1000	0.08~2m	0.073
382	UCAS_AOD	2014	OD	2	RGB	976	0~1	—	3.24
383	PKLot	2015	OD	1	RGB	12.4K	1280	—	4.6
384	CARPK	2016	OD	1	RGB	1.4K	1280	—	2
385	COWC	2016	OD	1	RGB	388.4K	256	0.15m	64
386	DLR3k/DLR-MVDA	2016	OD	7	RGB	20	3744	0.13m	0.162
387	DSTL16B	2016	OD	10	MS (Satellite)	1.4K	3391	—	7.3
388	DSTL3B	2016	OD	10	OPT (Satellite)	450	3391	—	12.9
389	HSI Vehicle	2016	OD	1	HS (Aerial)	157	—	—	—
390	LEVIR	2016	OD	3	RGB	22K	800	0.5m	—
391	Stanford Drone Dataset	2016	OD	6	RGB Video	—	—	0.025m	69
392	HRSC2016	2017	OD	26	RGB	1.1K	1100	0.4~2m	8
393	Land Covers Dstl Satellite Imagery	2017	OD	10	RGB+MS	57	3~348	0.3m-7.5m	21.7
394	NOAA Sea Lion	2017	OD,C	4	RGB	950	4900	—	96
395	OpenSARShip	2017	OD	17	Sentinel-1	11.3K	—	10m	0.87
396	OpenSARShip 2.0	2017	OD	17	Sentinel-1	34.5K	120	10m	0.87
397	PlanesNet	2017	OD	—	OPT (Satellite)	32K	20	—	0.39
398	RSOD	2017	OD	4	RGB	946	1000	0.3~3m	0.309
399	SSDD	2017	OD	1	SAR (Satellite)	1.2K	500	1~15m	0.995
400	WAMI DIRSIG	2017	OD	1	HS (Aerial)	55.2K	64	0.3m	0.327
401	Airbus Ship Detection Challenge	2018	OD	—	RGB	192.6K	768	—	30
402	DOTA	2018	OD	15	RGB	5.2K	800~4K	—	—
403	DOTA v1.0	2018	OD	15	RGB	2.8K	4000	—	18
404	EU Flood	2018	OD	4	OPT (Multiple)	3.7K	1280	—	5.6
405	HighD	2018	OD	2	RGB	110K	4096×2160	—	—
406	Humpback Whale Identification	2018	OD,C	—	OPT (Multiple)	25.5K	1050	—	0.7
407	Iceberg Detection	2018	OD	—	SAR (Satellite)	10K	75	—	0.29
408	ITCVD	2018	OD	1	RGB	173	5616	0.1m	1.3
409	Map Challenge	2018	OD	2	RGB	341.1K	300	—	—
410	Ships in Google Earth	2018	OD	2	RGB	794	2000	—	2
411	SpaceNet-4	2018	OD	—	MS+WorldView2	60K	900	0.3m	186
412	TBF	2018	OD	2	RGB	13	0~40	—	—
413	WHU Building	2018	OD	1	RGB	25.6K	512	0.3m	25
414	AIR-SARShip-1.0	2019	OD	1	SAR (Satellite)	31	3000	1~3m	0.2
415	ALCD	2019	OD	—	MS+S2	38	1830	10~60m	0.234
416	Bridges	2019	OD	1	RGB	500	3822	0.5m	1.45
417	built-structure-count	2019	OD	—	RGB	5.4K	512	0.3m	2
418	DLR-ACD	2019	OD	1	OPT (Aerial)	33	4458	—	—
419	DOTA v1.5	2019	OD	16	RGB	2.8K	4000	—	18
420	Interaction Dataset	2019	OD	1	RGB	10.9K	—	—	—
421	Microsoft Building (Australia)	2019	OD	—	—	11.3M	—	—	6.4
422	MTARSI	2019	OD	1	RGB	9.4K	256	—	0.48
423	NYCPlanimetric	2019	OD	24	OPT (Aerial)	2.1K	5000	—	2
424	Oil Palm Plant	2019	OD	2	RGB	20K	—	3m	0.5
425	Oil Storage Tanks	2019	OD	1	RGB	10K	512	0.5m	3
426	SAR Ship Detection	2019	OD	1	SAR	43.8K	256	3m-10m	0.407
427	SAR VD	2019	OD	1	SAR (Aerial)	104	980	—	7.5
428	SIMD	2019	OD	15	RGB	5K	1024	—	1
429	Swimming pool/car	2019	OD	2	RGB	3.8K	224	—	0.12
430	VisDrone2019-DET	2019	OD	10	RGB	10.2K	2000×1500	—	2
431	VisDrone2019-MOT	2019	OD	5	RGB Video	40K	3840×2160	—	14
432	VisDrone2019-SOT	2019	OD	3	RGB Video	139.3K	3840×2160	—	68
433	VisDrone2019-VID	2019	OD	5	RGB	40K	3840×2160	—	14
434	Aerial Maritime Drone	2020	OD	5	RGB	508	800×600	—	0.038
435	Agriculture-Vision	2020	OD	6	HS (Aerial)	95K	512	0.463m	4.39
436	AIR-SARShip-2	2020	OD	1	SAR (Satellite)	300	1000	1~3m	0.224
437	AI-TOD	2020	OD	—	RGB	28K	—	0.3m-30m	22
438	AU-AIR	2020	OD	8	RGB	32.8K	1920	—	2.2
439	BIRDSAI	2020	OD	—	TIR	162K	640	—	43.2
440	DIOR	2020	OD	20	RGB	23.5K	800	0.5m-30m	—
441	DOTA v2.0	2020	OD	18	RGB	11.3K	4000	0.1~0.81	34.3
442	DroneRGBT	2020	OD	1	Multiple (Drone)	3.6K	640	—	—
443	EAGLE	2020	OD	—	RGB	8.8K	936	0.05~0.45m	—
444	EoC	2020	OD	14	OPT (Aerial)	2.8K	256	—	0.217
445	FAIR1M	2020	OD	37	OPT (Satellite)	15.3K	5000	—	120
446	FGSD	2020	OD	43	OPT (Satellite)	4.7K	930	—	—
447	FUSAR-Ship	2020	OD,C	113	SAR (Satellite)	5K	512	—	—
448	HRSID	2020	OD	1	S1+SAR	5.6K	800	0.5~3m	0.581
449	IDTreeS	2020	OD,C	33	Multiple (Multiple)	591×369	20	—	0.28
450	Intersection Drone	2020	OD	5	RGB	11.5K	4096×2160	—	—
451	LS-SSDD	2020	OD	1	SAR (Satellite)	15	20000	1~15m	7.8

Continued on next page

Table 1 continued

Index	Dataset	Year	Task	# Cls.	Sensor / modality	# Samples	Size (px)	Resolution (m)	GB
452	MOR-UAV	2020	OD	2	RGB	10.9K	608	–	–
453	NEON Tree Crowns	2020	OD	2	RGB	11K	–	–	27.4
454	Oil and Gas Tank (OGST)	2020	OD	2	RGB	10K	512	0.3m	1.87
455	RarePlanes	2020	OD	110	MS+WorldView3	713.3K	512	0.3~1.5m	318
456	Roundabouts Drone	2020	OD	8	RGB	13.7K	4096x2160	–	–
457	RSOC	2020	OD	4	RGB	3.1K	2500	–	–
458	TGRS-HRRSD	2020	OD	13	RGB	21.8K	930	0.15m-1.2m	9
459	VALID	2020	OD,S	30	RGBD	6.7K	1024	–	15.7
460	VisDrone	2020	OD	–	RGB	275.4K	1400	–	85
461	WHU Cloud	2020	OD,S	1	RGB	859	512	30m	3.56
462	AFO	2021	OD	2	RGB	3.6K	3840	–	5
463	Airbus Aircraft Detection	2021	OD	1	RGB	103	2560	0.5m	0.092
464	Airbus Oil Storage Detection	2021	OD	1	MS	98	2560	1.2m	0.101
465	AMD	2021	OD	5	OPT (Drone)	508	800	–	0.037
466	AnimalDrone	2021	OD	–	OPT (Drone)	53.6K	–	–	–
467	ArcticSeals	2021	OD	1	MS (Aerial)	88K	6576	–	1K
468	ArtifiVe-Potsdam	2021	OD	1	MS	4.8K	600	0.05m	15.6
469	CASIA-aircraft	2021	OD	2	RGB	58.1K	399	–	–
470	CASIA-Ship	2021	OD	2	RGB	1.1K	1~680	–	–
471	CGI Planes	2021	OD	2	RGB	500	–	–	0.7
472	DroneVehicle	2021	OD	5	RGB-Infrared	56.9K	840	–	–
473	DSSDD	2021	OD	1	SAR (Satellite)	1.2K	256	–	1.27
474	Flying Airplanes	2021	OD	1	OPT (Satellite)	8K	–	–	491
475	Forest Damage	2021	OD	5	RGB	1.5K	1500	–	3.3
476	Forest Damages – Larch Casebearer	2021	OD,C	4	RGB	1.5K	1500	–	3.3
477	LEVIR-Ship	2021	OD	1	OPT (Satellite)	3.9K	512	–	0.821
478	MultiDrone	2021	OD	9	OPT (Drone)	259.8K	1280	–	–
479	NASADEbris	2021	OD	1	OPT (Satellite)	707	256	–	0.073
480	NEON Tree Dataset	2021	OD	1	Multiple (Aerial)	2.2K×426	40	–	8.4
481	SaRNet	2021	OD	1	OPT (Satellite)	2.6K	1000	–	0.894
482	SeabirdsDetection	2021	OD	1	OPT (Aerial)	1	28814	–	2.2
483	Search and Rescue (SaR)	2021	OD	1	RGB	2.6K	1000	0.5m	–
484	ShipRSImageNet	2021	OD	1	OPT (Satellite)	3.4K	930	–	4.3
485	SRSDD-v1.0	2021	OD	6	SAR (Satellite)	666	1024	–	0.713
486	SWIM	2021	OD	1	RGB	14.6K	768	0.5~2.5m	12
487	TopoBoundary	2021	OD	8	OPT (Aerial)	25.3K	1000	–	20
488	Urban Tree Crown Detection	2021	OD	1	OPT (Aerial)	220	512	–	0.2
489	WHUS2-CD+	2021	OD	–	S2	36	10980	10m	27.8
490	xView3-SAR	2021	OD	2	S1	991×2	27000	10m	2K
491	Yosemite Tree	2021	OD	1	OPT (Aerial)	1	38400	–	2.7
492	AISD	2022	OD	–	RGB	–	–	–	–
493	BONAI	2022	OD	1	RGB	3.3K	1~24	0.3m-0.6m	4.86
494	Buildings Containers Cranes	2022	OD	3	OPT (Satellite)	–	2500	–	3
495	DroneCrowd	2022	OD	1	OPT (Drone)	33.6K	1920	–	10.4
496	HRPlanesv2	2022	OD	1	OPT (Satellite)	2.1K	4800	–	8.1
497	MSAR	2022	OD	4	SAR (Satellite)	28.4K	256	–	2.5
498	OnlyPlanes	2022	OD	1	OPT (Satellite)	12.5K	1024	–	15
499	RampBuilding	2022	OD	1	OPT (Satellite)	96.3K	256	–	24.8
500	ReforesTree	2022	OD,C	6	OPT (Drone)	100	4000	–	7.5
501	RSDD-SAR	2022	OD	–	SAR (Satellite)	7K	512	–	–
502	SADD	2022	OD	1	SAR (Satellite)	3K	224	–	–
503	SAR Aircraft	2022	OD	2	SAR	3K	224	0.5m-3m	0.18
504	Satellite Pool Detection	2022	OD	1	OPT (Satellite)	14.9K	512	–	3.7
505	SeaDronesSee	2022	OD	6	RGB	5.6K	3~840	–	60.3
506	SODA	2022	OD	9	RGB	2.5K	2.7K~4.8K	–	–
507	UAVOD-10	2022	OD	10	OPT	844	–	–	–
508	VISO	2022	OD	4	OPT (Satellite)	15.5K	12000	–	8.6
509	MDMT	2023	OD	–	OPT (Drone)	39.7K	–	–	25.8
510	EverWatch	2024	OD	8	RGB	5.3K	1500	0.01m	–
511	LEVIRDet-159	2026	OD	159	OPT	700K	–	–	–
512	MSDS900	–	OD	1	OPT (Satellite)	3K	900	–	–
513	MSDSfull	–	OD	1	OPT (Satellite)	200	4800	–	–
514	PSDS	–	OD	1	OPT (Satellite)	1.3K	900	–	–
515	SMCDD	–	OD	–	SAR (Satellite)	–	–	–	–
516	Sydney caption	2016	IC	7	RGB	613	500	0.5m	0.298
517	UCM caption	2016	IC	21	RGB	2.1K	256	0.3m	0.255
518	RSICD	2017	IC	–	OPT+Text	10.9K	–	–	–
519	RSITMD	2021	IC	–	OPT+Text	4.7K	–	–	–
520	LEVIR-CC	2022	IC	–	RGB	10.1K	512	–	–
521	ChatEarthNet	2024	IC	–	S2+Text	173.5K	256	10m	38.4
522	LHRS-Align	2024	IC	–	RGB	–	–	0.5m-30m	–
523	VRSBench	2024	IC	–	OPT+Text	29.6K	–	–	–
524	RSCC	2025	IC	–	OPT+Text	62.4K	–	–	–
525	BigEarthNet.txt	2026	IC	–	S1/S2+Text	464K	–	10m	–
526	Sentinel2Cap	2026	IC	–	S2+Text	–	–	–	–
527	DFC06	2006	SR	–	MS (Aerial)	6	5000	–	0.39
528	Proba-V Super Resolution	2018	SR	–	MS (Satellite)	1.2K	384	–	0.692
529	ALSAT-2B	2021	SR	–	OPT (Satellite)	5.5K	256	–	0.07
530	PAirMAX (Airbus)	2021	SR	–	MS (Satellite)	14	–	–	0.153
531	PAirMAX (MAXAR)	2021	SR	–	OPT	14	–	–	–
532	PushFrameSR	2021	SR	–	OPT (Satellite)	960×15	256	–	1.1
533	Jilin1 VidSR	2022	SR	–	OPT (Satellite)	6.2K	3000	–	19.4
534	MultiExposureSR	2022	SR	–	OPT (Satellite)	–	–	–	0.743
535	Mu-S2	2022	SR	–	MS (Satellite)	1.3K	100	–	2.8
536	PRISMA Pan Sharp	2022	SR	–	HS (Satellite)	4	2400	–	1.7
537	SEN2VENμS	2022	SR	–	S2+VENμS	133K	256	10m-20m	85
538	WorldStrat	2022	SR	8	Multi	3.4K	1054	1.5m-60m	110
539	Africa MiniCubes	2021	P	–	Multi	–	–	–	–
540	SeCo	2021	P	–	S2	200K×5	264	10m	43.6
541	Hephaestus	2022	P	–	SAR	110.6K	–	–	–
542	SSL4EO-S12	2022	P	–	S1/S2	251.1K	–	–	–

Continued on next page

Table 1 continued

Index	Dataset	Year	Task	# Cls.	Sensor / modality	# Samples	Size (px)	Resolution (m)	GB
543	Clay v1.5 pretraining corpus	2024	P	–	Multi	70M	156x256	0.5m-500m	–
544	DOFA-data (Metadataset)	2024	P	–	Multi	8.1M	–	0.5m-30m	12K
545	EarthDial	2024	P	–	Multi	11.1M	–	–	–
546	GRAFT	2024	P	–	RGB+MS	–	–	–	–
547	M3LEO	2024	P,C	–	S1/S2	10M	–	10m	17.5K
548	MajorTOM-Core	2024	P	–	S1/S2+DEM	1.5M~2.2M	356~1.1K	10m~30m	63K
549	MMEarth	2024	P	–	Multi	1.2M	128	10m	597
550	Prithvi-EO-2.0 pretraining corpus	2024	P	–	HLS (L30/S30)	4.2M	256	30m	–
551	Satelloic (EarthView)	2024	P	–	Multi	–	–	1m-40m	–
552	SkyScript	2024	P	–	RGB+Text	5.2M	–	0.1m-30m	–
553	SkySense	2024	P	–	Multi	–	–	0.31m-20m	–
554	SpectralEarth	2024	P	–	EnMAP HS	539K	128	30m	3.3K
555	Copernicus-Pretrain	2025	P	–	Multi	18.7M	–	–	–
556	GeoPlex	2025	P	–	Multi	5 datasets	–	0.2m-250m	–
557	SAR-GEOVL-1M	2025	P	–	SAR	120K	–	–	–
558	SkySense++ pretraining corpus	2025	P	–	Multi	27M	–	0.5m-30m	–
559	SSL4EO-S12 v1.1	2025	P	–	S1/S2	246.1K×4	–	10m	–
560	TerraMesh	2025	P	–	Multi	9M	264	10m-30m	31K
561	SpectralEarth-MM	2026	P	–	Multi	2M loc.	–	–	40K
562	Copernicus-Bench	2025	B	–	Multi	15 datasets	–	–	–
563	PlantationBench	2025	B	–	Multi	–	–	–	–
564	EarthShift	2026	B	–	Multi	11 tasks	–	–	–
565	MMEarth-Bench	2026	B	–	Multi	5 tasks	–	–	–

C = classification, R = regression, S = semantic segmentation, I = instance segmentation, T = time series, CD = change detection, OD = object detection, IC = image captioning, SR = super resolution, P = pretraining, B = benchmark suite. K = thousand, M = million, B = billion. Long multi-sensor combinations are shown as *Multi*. Embedding-only datasets and 3D / point-cloud datasets are excluded. Dataset names are clickable external links. EarthNets citations are linked to the cited paper when the citation is suitable; explicit arXiv identifiers are linked directly. Where EarthNets does not expose a stable paper identifier, the link is an exact Google Scholar citation query rather than a guessed DOI. Ambiguous, non-paper, or mismatched references use a dataset/source page or an exact-name external search.

The inventory presents a compilation of Earth Observation datasets identified across the literature and public repositories. Rather than being restricted exclusively to downstream evaluation benchmarks, it encompasses a broad spectrum of resources, including large-scale pre-training archives and specialized domain-specific collections. Direct hyperlinks are provided in place of formal bibliographic entries to streamline tabular navigation.

References

- [1] Isaac Corley et al. *No One Knows the State of the Art in Geospatial Foundation Models*. 2026. arXiv: 2605.12678 [cs.CV]. URL: <https://arxiv.org/abs/2605.12678>.
- [2] Patrick Helber et al. “EuroSAT: A Novel Dataset and Deep Learning Benchmark for Land Use and Land Cover Classification”. In: *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing* 12.7 (2019), pp. 2217–2226. DOI: 10.1109/JSTARS.2019.2918242.
- [3] Gong Cheng, Junwei Han, and Xiaoqiang Lu. “Remote Sensing Image Scene Classification: Benchmark and State of the Art”. In: *Proceedings of the IEEE* 105.10 (2017), pp. 1865–1883. DOI: 10.1109/JPROC.2017.2675998.
- [4] Gui-Song Xia et al. “AID: A Benchmark Data Set for Performance Evaluation of Aerial Scene Classification”. In: *IEEE Transactions on Geoscience and Remote Sensing* 55.7 (2017), pp. 3965–3981. DOI: 10.1109/TGRS.2017.2685945.
- [5] Colorado J. Reed et al. “Scale-MAE: A Scale-Aware Masked Autoencoder for Multiscale Geospatial Representation Learning”. In: *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 2023, pp. 4088–4099. DOI: 10.1109/ICCV51070.2023.00378.
- [6] Alexandre Lacoste et al. *GEO-Bench: Toward Foundation Models for Earth Monitoring*. 2023. arXiv: 2306.03831 [cs.LG].
- [7] Valerio Marsocci et al. *PANGAEA: A Global and Inclusive Benchmark for Geospatial Foundation Models*. 2024. arXiv: 2412.04204 [cs.CV].
- [8] Adam J. Stewart et al. “TorchGeo: Deep Learning With Geospatial Data”. In: *ACM Transactions on Spatial Algorithms and Systems* 11.4 (2025), pp. 1–28. DOI: 10.1145/3707459.
- [9] Nicolas Karasiak et al. “Spatial Dependence Between Training and Test Sets: Another Pitfall of Classification Accuracy Assessment in Remote Sensing”. In: *Machine Learning* 111.7 (2022), pp. 2715–2740. DOI: 10.1007/s10994-021-05972-1.
- [10] Teja Kattenborn et al. “Spatially Autocorrelated Training and Validation Samples Inflate Performance Assessment of Convolutional Neural Networks”. In: *ISPRS Open Journal of Photogrammetry and Remote Sensing* 5 (2022), p. 100018. DOI: 10.1016/j.ophoto.2022.100018.
- [11] Isaac Corley et al. “Revisiting Pre-trained Remote Sensing Model Benchmarks: Resizing and Normalization Matters”. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*. 2024, pp. 3162–3172. DOI: 10.1109/CVPRW63382.2024.00322.
- [12] Amandeep Kaur et al. “Pretrain Where? Investigating How Pretraining Data Diversity Impacts Geospatial Foundation Model Performance”. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*. 2026, pp. 7949–7959.
- [13] Michael Schmitt et al. “There Are No Data Like More Data: Datasets for Deep Learning in Earth Observation”. In: *IEEE Geoscience and Remote Sensing Magazine* 11.3 (2023), pp. 63–97. DOI: 10.1109/MGRS.2023.3293459.

- [14] IEEE Geoscience and Remote Sensing Society. *Earth Observation Database*. URL: <https://eod-grss-ieee.com/> (visited on 09/15/2026).
- [15] Zhitong Xiong et al. “EarthNets: Empowering AI in Earth Observation”. In: *arXiv preprint arXiv:2210.04936* (2022). DOI: 10.48550/arXiv.2210.04936. arXiv: 2210.04936.
- [16] Christoph Wiedemann. *Awesome Satellite Imagery Datasets*. Archived community-maintained collection of annotated satellite and aerial imagery datasets. URL: <https://github.com/chrieke/awesome-satellite-imagery-datasets> (visited on 09/15/2026).
- [17] Satellite Image Deep Learning. *Datasets for Deep Learning with Satellite and Aerial Imagery*. URL: <https://github.com/satellite-image-deep-learning/datasets> (visited on 09/15/2026).